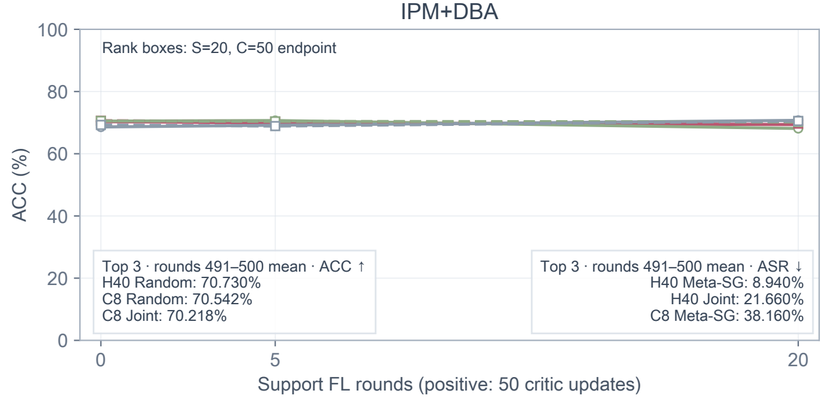
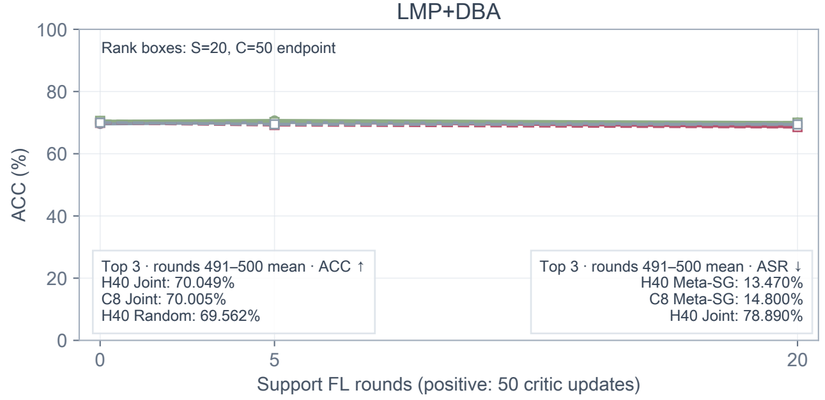
Meta-SG: shared-support sample budgets

CIFAR-10 / CIFAR10CNN / IID / 20 clients, all participate / 4 malicious (2 + 2) / train seed 42, query seed 43

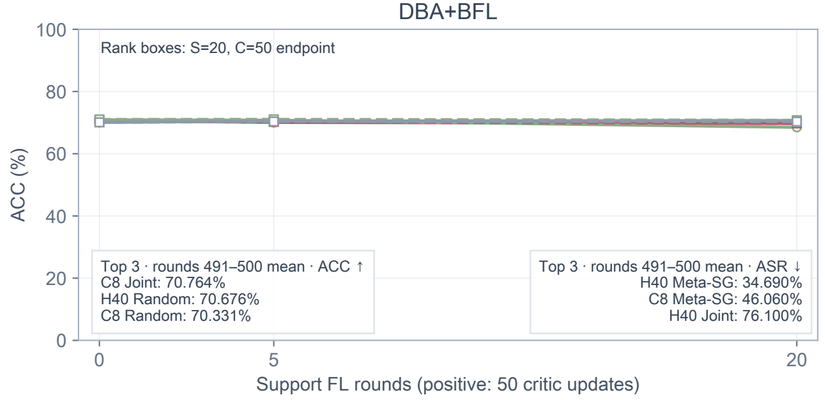
IPM+DBA



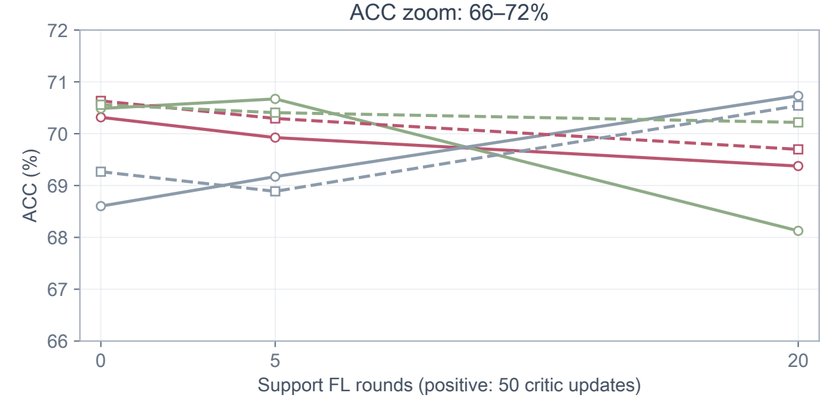
LMP+DBA



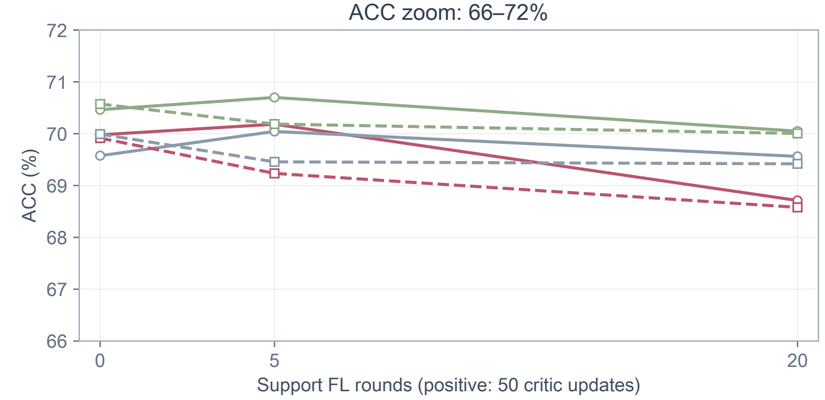
DBA+BFL



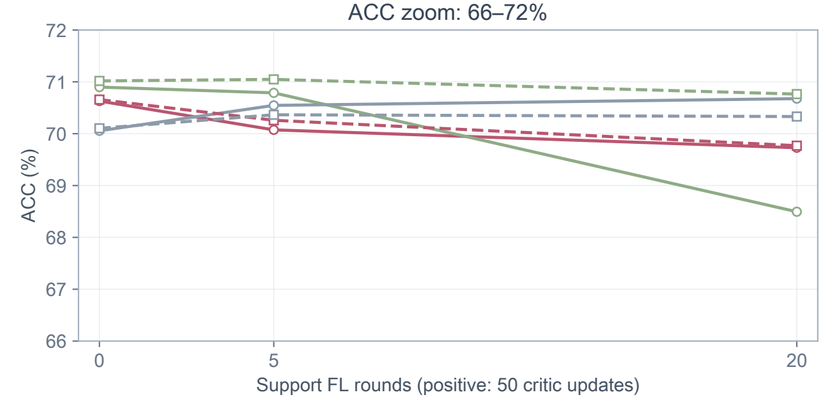
ACC zoom: 66–72%



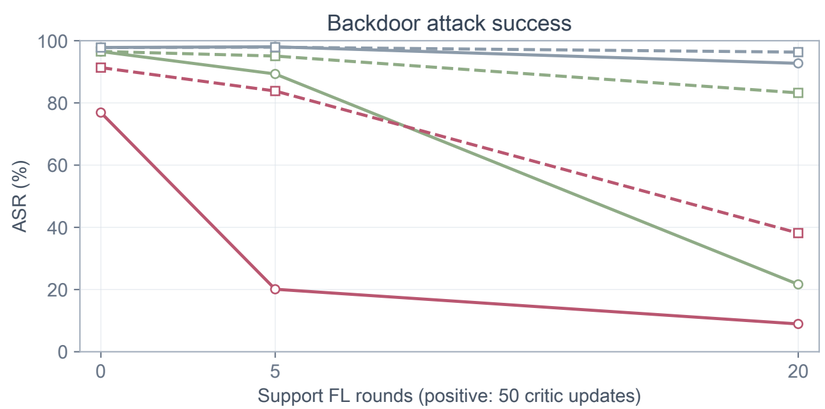
ACC zoom: 66–72%



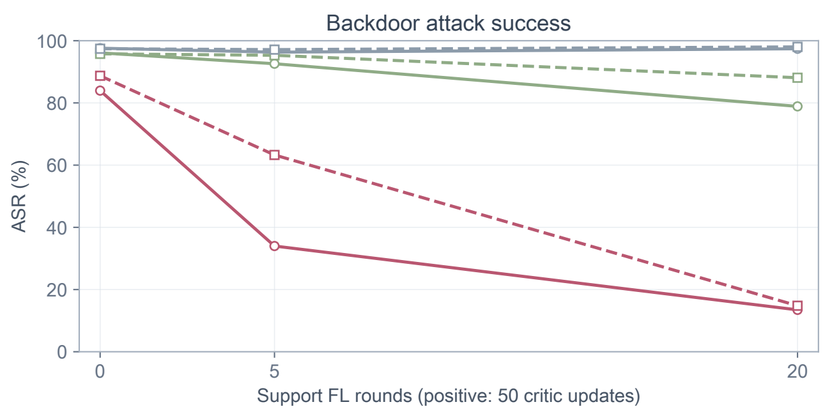
ACC zoom: 66–72%



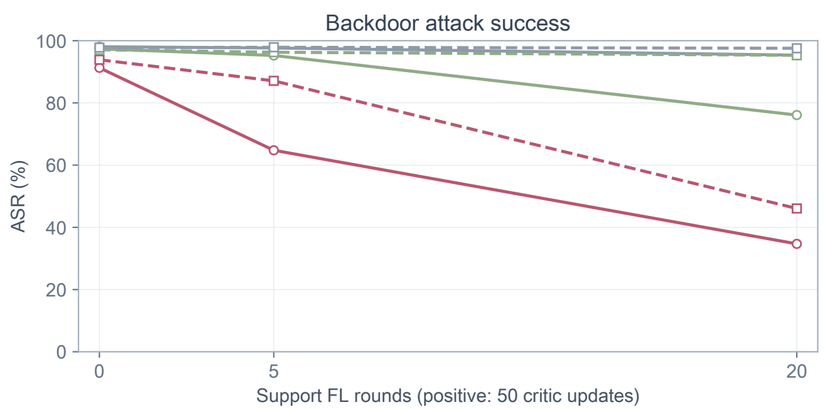
Backdoor attack success



Backdoor attack success



Backdoor attack success



H40 Meta-SG · H40 Random · C8 Joint · H40 Joint · C8 Meta-SG · C8 Random

Each point: independent 500-round frozen query, rounds 491–500 mean. Zero = no support and no updates. One seed; no smoothing or error bars.
Shared uniform-action support per scenario. Positive budgets: base and extra mean-Q actor updates each = half the critic updates.
Local SGD: 5 steps, batch 128, learning rate 0.05. H40 solid circles; C8 dashed squares. Training numerical terminations differ.